

Physical Constants as Middle-Way Fixed Points: A Dynamical Resolution of the Fundamental Constant Problem

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Abstract

Modern physics contains a number of unexplained empirical constants, including the speed of light c , the Planck constant h , the gravitational constant G , the cosmological constant Λ , and various coupling constants. These constants are treated as fixed, universal quantities, yet no fully satisfactory explanation has been given for why they take their observed values.

This paper proposes a new approach: physical constants are not static, primitive entities, but are instead *Middle-Way Fixed Points* that arise from the dynamical interaction of uncertainty (emptiness), stochastic proposals (ranki), and structural constraints (dependent origination). These components form a general coherence functional whose minima define the apparent “constants.” The resulting framework resolves the Fundamental Constant Problem by replacing static equality with a dynamic equivalence operator, written $\overset{\text{mw}}{=}$, which captures convergence to minimal-coherence fixed points rather than strict identity.

This view parallels the resolution of classical logical paradoxes (Godel incompleteness) and computational barriers (such as the P vs NP problem in the companion paper “P $\overset{\text{mw}}{=}$ NP”).

1 Introduction

Why do the constants of nature take the values they do? The question is older than relativity or quantum theory, and persists across modern physics. Despite impressive theoretical developments, no derivation exists for the numerical values of c , h , G , Λ , or the masses of elementary particles.

Traditionally, physics assumes a fixed observer frame and a fixed background structure. Under these assumptions, constants necessarily appear arbitrary. This mirrors earlier difficulties in logic and computation, where static frameworks produced undecidability and incompleteness phenomena.

We propose that the static assumption itself is the root problem. If the observer is not fixed—that is, if the measurement frame is part of the dynamical system—then constants are not primitive but emergent. They arise as stable fixed points in a coherence-minimizing process.

2 Dynamic Structure: Emptiness, Ranki, and Dependent Origination

We model the emergence of physical constants by three structural components:

1. **Emptiness (E)**: an initial space of uncertainty, representing the degrees of freedom in the measurement and physical configuration.

2. **Ranki (R)**: stochastic proposals generated from E , analogous to non-deterministic sampling, evolutionary search, or Monte Carlo exploration.
3. **Dependent Origination (D)**: a constraint functional that assigns coherence energy to each proposal.

Let x be a candidate value (e.g., for c or G). Define the coherence energy functional:

$$E(x) = SC(x) + CL(x) + OD(x),$$

where SC (self-consistency), CL (cross-linearity), and OD (observer-dependence) are non-negative terms.

A physical constant is then defined not by a static equation, but as a *minimal-coherence fixed point*:

$$x^* = \arg \min_x E(x).$$

3 The Middle-Way Equivalence Operator

The traditional equality sign is inadequate for dynamical systems in which the observer and the observed mutually constrain each other. We therefore introduce the Middle-Way equivalence operator:

$$A \stackrel{\text{mw}}{=} B \quad \text{iff} \quad A \text{ and } B$$

converge to the same minimal-coherence fixed point.

This operator generalizes classical equality by grounding equivalence in dynamical convergence rather than static identity.

Applying $\stackrel{\text{mw}}{=}$ to physical constants means that “ c is constant” is not a statement of rigid identity, but of stability under dynamical observation. This resolves longstanding confusions about whether c “really” changes or is an illusion of measurement: both sides converge in the limit to the same coherence minimum.

4 Application to the Speed of Light

The speed of light has long been a paradoxical quantity. It is invariant across observers, yet emerges from the electromagnetic properties of the vacuum:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.$$

Under our framework, this is natural. c is the minimal-coherence fixed point of:

- the structure of spacetime geometry,
- electromagnetic propagation,
- information-theoretic constraints on causality,

- and the dynamical observer configuration.

Thus:

$$c \stackrel{\text{mw}}{=} x^*,$$

and the speed of light appears constant because it is the stable equilibrium of the coherence functional.

5 Resolution of the Fundamental Constant Problem

The proposed framework resolves the Fundamental Constant Problem without fine-tuning or anthropic reasoning:

1. Constants are not primitive, but emergent.
2. Their values are determined by coherence minimization.
3. The observer is part of the system, so constants must be understood dynamically.
4. Classical equality is replaced by Middle-Way equivalence: $A \stackrel{\text{mw}}{=} B$.

In this view, physical constants are not arbitrary. They are the stable fixed points of a deep dynamical structure.

6 Conclusion

We have proposed that physical constants arise as Middle-Way fixed points—minimal-coherence equilibria determined by the interaction of uncertainty, stochastic proposals, and structural constraints. This approach replaces static equality with a dynamic equivalence relation and resolves longstanding foundational puzzles.

This framework parallels similar resolutions in logic and computation, suggesting a unified dynamical architecture for emergent structure across scientific domains.